

# **Capstone Project - Development of a Web-Based Knowledge Repository for Analytical Geoscience Workflows**

Group 4

# Table of Contents

|                                                     |           |
|-----------------------------------------------------|-----------|
| <b>1.1 Introduction.....</b>                        | <b>3</b>  |
| <b>2.1 Methods.....</b>                             | <b>3</b>  |
| 2.1.1 Knowledge Repository.....                     | 3         |
| 2.1.2 Interactive Data Atlas.....                   | 4         |
| 2.1.3 SEM Image Metadata Tool:.....                 | 5         |
| 2.1.4 SEM Scale Prediction Model.....               | 5         |
| <b>2.2 Results.....</b>                             | <b>7</b>  |
| 2.2.1 Knowledge Repository.....                     | 7         |
| 2.2.2 Interactive Data Atlas.....                   | 8         |
| 2.2.3 SEM Image Metadata Tool.....                  | 10        |
| 2.2.4 SEM Scale Prediction Model.....               | 12        |
| <b>3.1 Discussion.....</b>                          | <b>12</b> |
| <b>4.1 Conclusion.....</b>                          | <b>13</b> |
| <b>5.1 References.....</b>                          | <b>15</b> |
| <b>6.1 Appendix.....</b>                            | <b>15</b> |
| Task allocation + Contributions of each member..... | 15        |
| Team Github Repository.....                         | 16        |

# 1.1 Introduction

Our clients, the Centre for Microscopy, Characterisation and Analysis (CMCA) at the University of Western Australia (UWA) supports critical minerals research using a wide range of microscopy and spectroscopy techniques. A key challenge the CMCA faces is the lack of structure, consistency and accessibility of complex analytical workflows, instrument output and other training resources, which are critical to performing and reproducing analyses. This introduces a bottleneck in research efficiency and increases the risk of damage to the equipment worth millions of dollars.

For this project, we were provided with geoscience data with no rigidly structure, including:

- Workflows, training documents and examples
- Textual sample information
- Sample images
- Instrument output images

Taking the data provided and the client requirements into account, we proposed a solution with the four main components below.

1. Knowledge Repository to host workflows and training resources
2. Interactive Data Atlas to visualize mineral samples
3. Metadata Extraction/Combination tool to obtain important experimental parameters/information
4. SEM Scale Prediction Model to predict missing information

These components are then linked together to form a comprehensive solution, attempting to maximise the usage and accessibility of all the unstructured data to help improve geoscience research efficiency and accessibility.

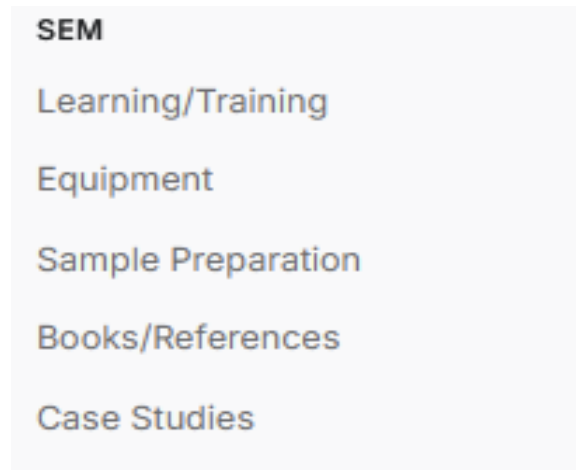
## 2.1 Methods

This section is divided into subsections, due to the substantial difference in nature of each component.

### 2.1.1 Knowledge Repository

The goal of the knowledge repository is to centralise research resources for accessibility and enable knowledge sharing. This meant that ease of use was also desirable. Taking this into account, Gitbook hosting was the proposed solution due to its professional but simple-to-use interface. GitBook supports embedded multimedia, as well as collaborative editing and version control with GitHub support. This meant that knowledge sharing was easily accessible, with version history protection.

The main approach for this component was designing a modular template, as the client requested modules to be constructed for each instrument type. These modules were then displayed using Cards on Gitbook. For each module, the following structure was used for the sub-pages:



**Figure 1: Sub-page Structure of Repository**

Once the overall structure was finalized, all the available documents and multimedia were organized into the corresponding module pages. Unsupported image formats such as .tif images were converted into .jpg to be displayed natively. The formatting was also standardized to make navigation simple and natural.

## 2.1.2 Interactive Data Atlas

While Gitbook provides an excellent platform for documentation and publishing documents, it does not provide a natural visualisation for geotagged information, where a map-based interface would be optimal. The solution could be decomposed into two main parts; developing a frontend map interface to display information, and a backend database to store all the sample information.

For the database, the choice of technology was PostgreSQL with PostGIS enabled. Discussions with CCRF researchers and our clients, as well as investigating the *Digital Atlas of Australia* [1] confirmed this as the standard implementation in geoscience. The inclusion of PostGIS also supports advanced spatial queries for future research..

The schema was then designed to maintain relations between samples, documents and instruments. This was done to display the instrument output images and reference documents related to the selected samples. Maintaining the relation for instruments also enables the addition of an equipment-layer filter for the map interface. Images were hosted remotely, with the links stored in the database.

Streamlit was chosen to develop the map interface. This was due to the compatibility with folium and streamlit-folium, enabling support for Leaflet maps and bidirectional data transfer. This

increased the interactivity between the interface and the backend database. SQLAlchemy was then used to connect the database to the streamlit application.

### 2.1.3 SEM Image Metadata Tool:

The Python-based metadata extraction and preprocessing tool was developed to address the challenges of managing Scanning Electron Microscope (SEM) data in research. SEM images often contain incomplete, inconsistent, or inaccessible metadata, making it difficult to reproduce experiments or compare results accurately. To overcome this, a secure, offline, and user-friendly Python application was created to efficiently extract, organize, and standardize SEM metadata while ensuring data privacy and reliability.

The tool automates metadata extraction and converts the results into structured formats such as JSON, image, and Excel files. It supports three main workflows:

1. **Extracting embedded metadata** directly from SEM images captured on instruments like the Helios NanoLab G3 CX, Supra 55VP, and VEGA3 XMU. Parameters such as accelerating voltage, magnification, and detector type are standardized into a unified format. This design was inspired by the *Microscopy Metadata Perspective* study [2], which highlights the value of structured metadata.
2. **Merging images with external metadata files**, where the tool automatically matches files based on names to consolidate information and improve consistency.
3. **Extracting SEM images and metadata from PDF publications**, enabling retrieval of both images and related text for use in machine learning or comparative analysis.

By operating entirely offline, the tool provides fast, secure processing. It bridges the gap between raw experimental data and its published representation, enhancing reproducibility, transparency, and data management in microscopy research.

### 2.1.4 SEM Scale Prediction Model

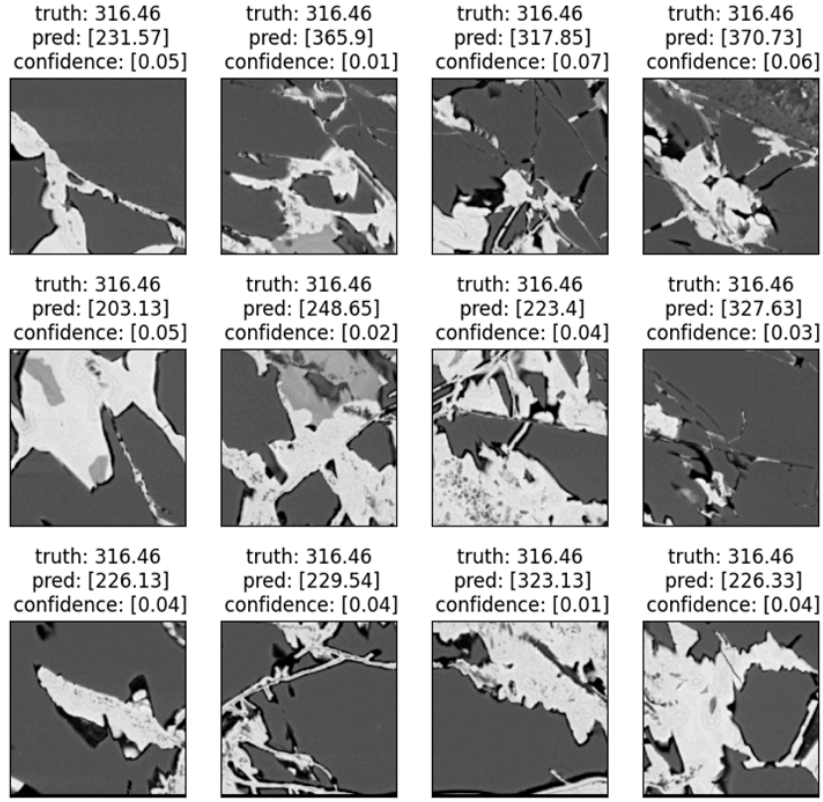
A problem with some instrument images is missing metadata due to image preprocessing. A potential method is to predict this information, specifically the scale information from SEM images from pixels with patterns. A neural network model is the ideal choice.

For this model, a Vision Transformer (ViT) [3] was chosen as the backbone. To predict high resolution images (which is not supported by ViT), large images were split into patches and used to train the model. After training, we evaluate the model with a validation set and calculate the confidence score with the equations below:

$$\text{inverse percentage error(IPE)} = 1 / \left( \frac{x_{pred} - x_{truth}}{x_{truth}} \right) \quad (1)$$

$$\text{confidence score} = (\text{IPE} - \text{Min}(\text{IPE})) / (\text{Max}(\text{IPE}) - \text{Min}(\text{IPE})) \quad (2)$$

Then, we modify the model (but keep the weights) and retrain it with the confidence score. When making predictions, we split large images into patches, predicting patches one by one, weighing predictions with confidence scores and calculating the weighted sum. The predicted value is pixel size (how many nm each pixel represents), which represents the scale.



**Figure 2: Example Predictions for Patches**

We implemented a U-Net diffusion pipeline (PyTorch, Hugging Face diffusers) to synthesise SEM-style images. Prior to modelling, images were standardised (filenames/metadata), resized to 256×256, and low-quality items removed. Trained on ~300 cleaned real images with a cosine schedule and Adam (1e-4), the model samples via iterative denoising (~1000 steps). The pipeline now produces realistic SEM-like textures suitable for downstream training.

## 2.2 Results

### 2.2.1 Knowledge Repository

The final knowledge repository provides a centralised, web-based platform that stores all analytical workflows and training materials in one location. Each module is structured to provide a seamless learning experience for geoscience researchers. As this component simply hosts materials, quantitative evaluation was not performed. Instead, we closely communicated with the client every week to ensure the implementation was done to the preference of our clients.

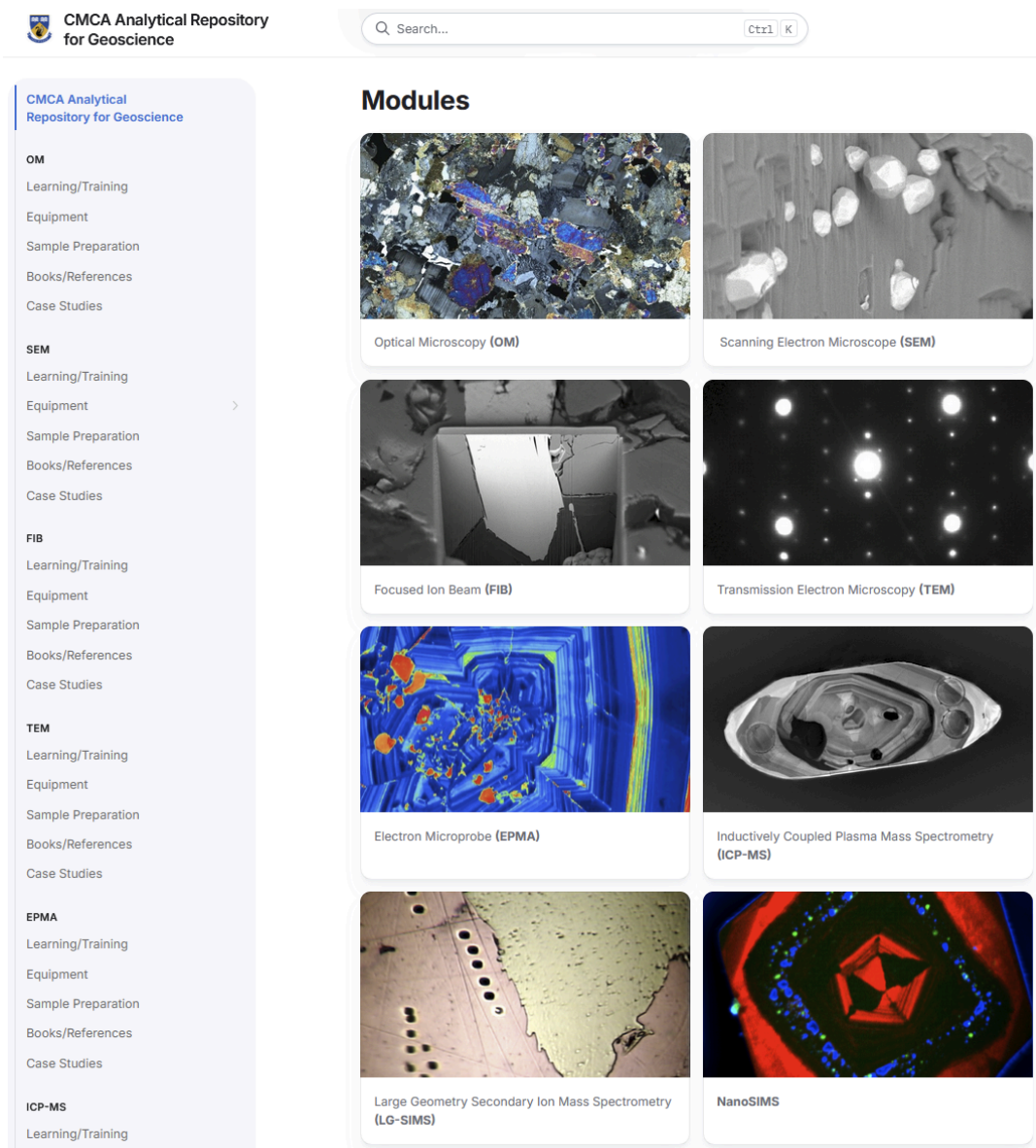


Figure 3: Knowledge Repository Homepage

## 2.2.2 Interactive Data Atlas

For this part, evaluation was performed using locally deployed versions due to resource constraints. This is still relatively accurate as the images were hosted online.

For queries of Samples, the time taken to display sample information never exceeded **500ms**. For the sample images, all images were displayed **within 2 seconds**, varying with image size. For the instrument images, most images were displayed **within 5 seconds**, only exceeding the threshold for very large images (> 500 MB). This delay mostly stems from the retrieval and loading time from the cloud-hosted images. Once the images are loaded, navigating to the original sample again will reload all images **within 1 second**, indicating that the bottleneck does not originate from the interface. The time taken to display References is **under 500ms**, as no image processing is required.

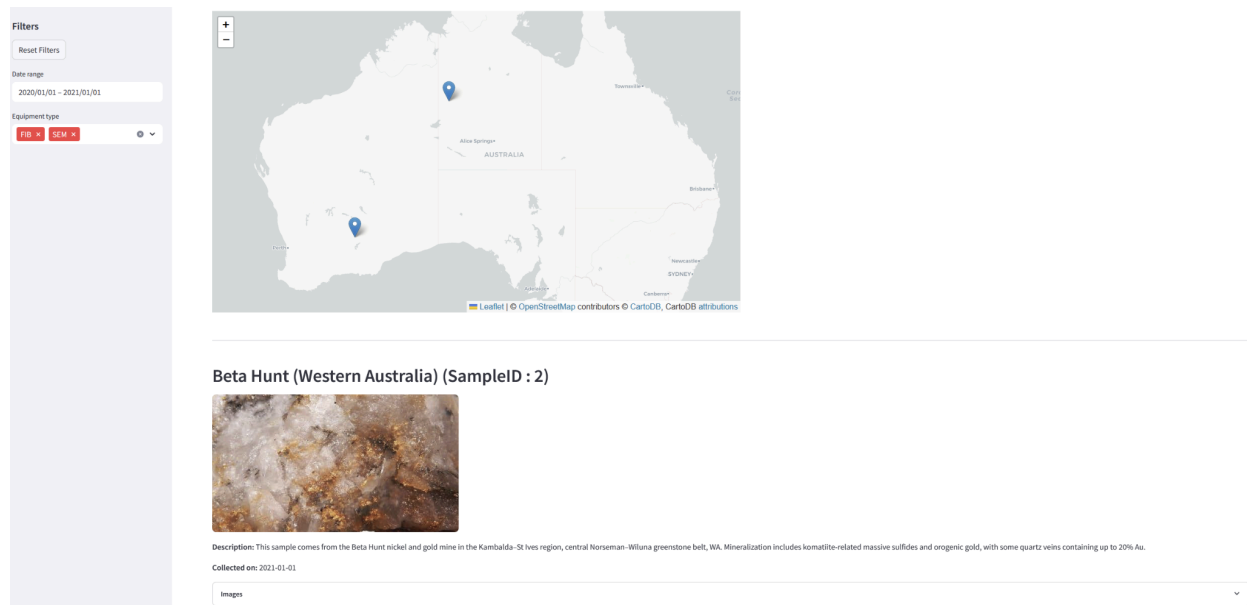
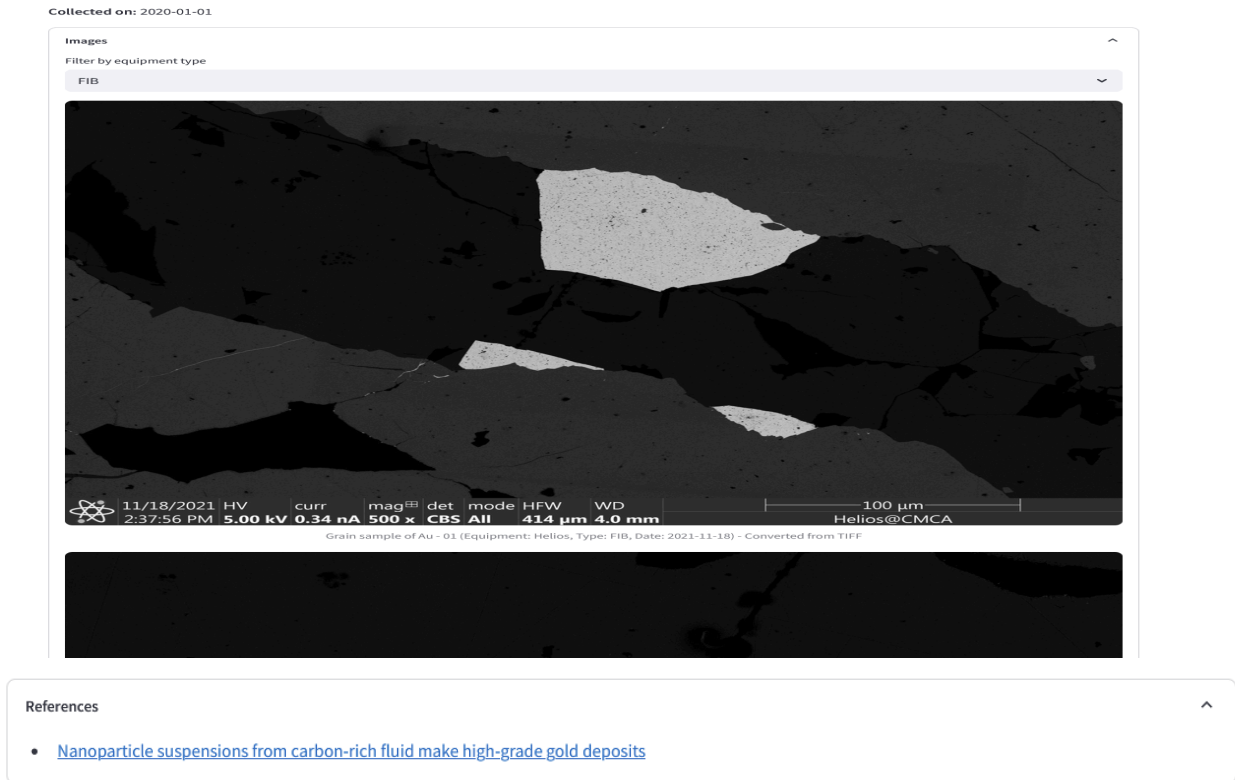


Figure 4: Atlas with Sample Information





**Figure 5: Instrument Image and Reference Tabs**

Both applying and resetting filters to the interface updates the map in **under 500ms**. The tested processing times were deemed reasonable for good user experience.

**Filters**

Reset Filters

Date range

2020/01/01 – 2021/01/01

Equipment type

FIB x SEM x x v

**Figure 6: Map Filters**

## 2.2.3 SEM Image Metadata Tool

The performance of the Python-based SEM metadata extraction tool was evaluated using datasets obtained from five different scanning electron microscopes: Helios NanoLab G3 CX, Supra 55VP, VEGA3 XMU, Verios 460, and Microscope Titan200.

Each dataset included images encoded with distinct metadata structures, ranging from binary data and embedded XML files to non-standard XML schemas. For embedded metadata extraction, the tool achieved an average accuracy of 96% in correctly identifying and parsing parameters such as accelerating voltage, magnification, working distance, and detector type.

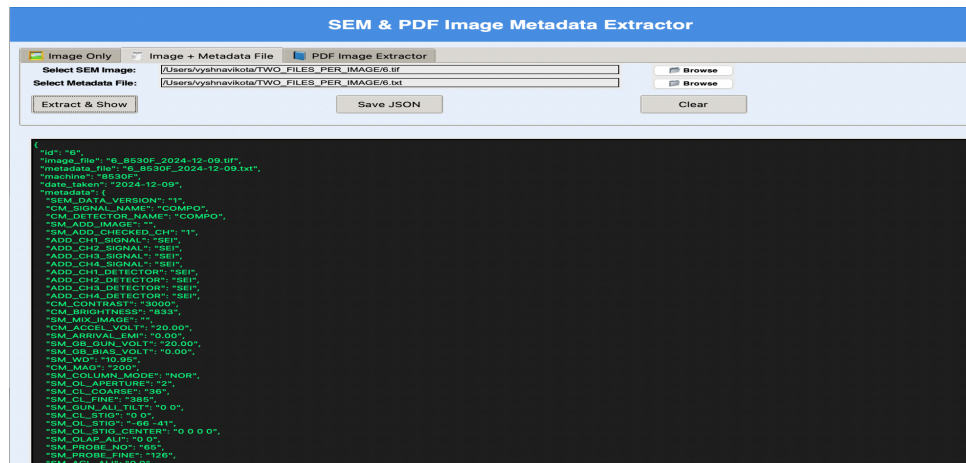


Figure 7: Embedded Metadata Extraction

In the second workflow, which merges SEM images with corresponding external metadata files, the matching accuracy reached 90%, even when filenames were partially inconsistent. This significantly reduced manual data organization time by more than 80% compared to traditional spreadsheet-based methods.

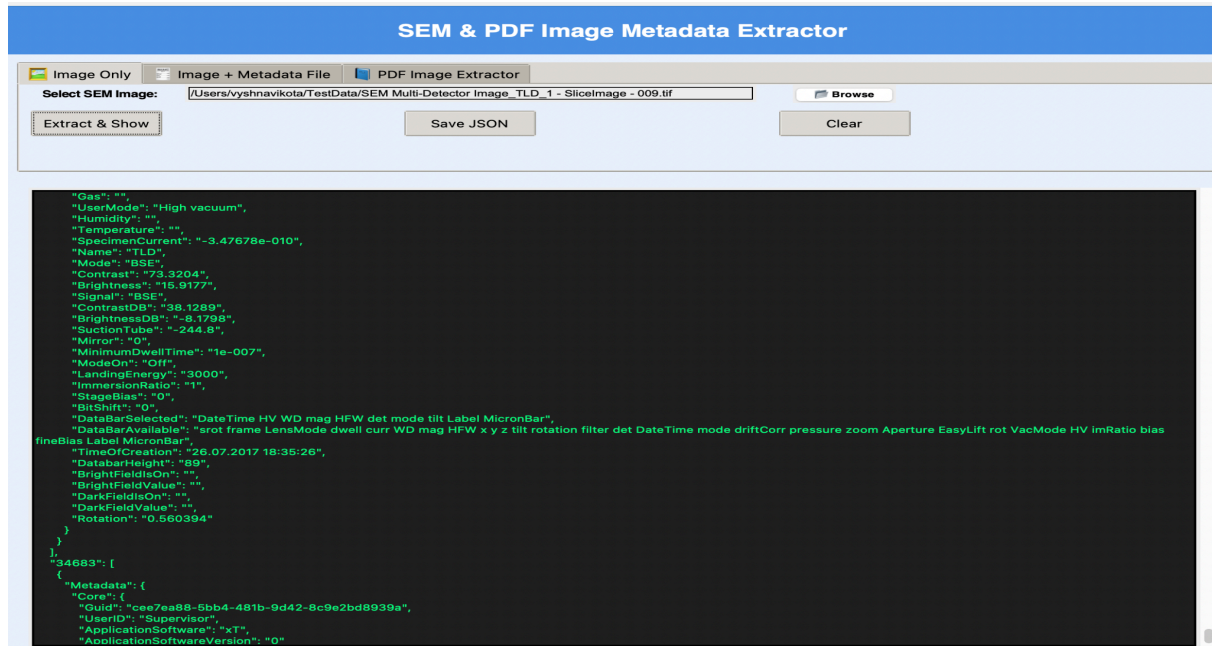


Figure 8: Image and Structured Metadata Merging

For the extraction of SEM images and associated data from published PDF documents, the tool successfully identified and extracted 90% of embedded figures and achieved 90% completeness in recovering related textual metadata, such as magnification and scale.

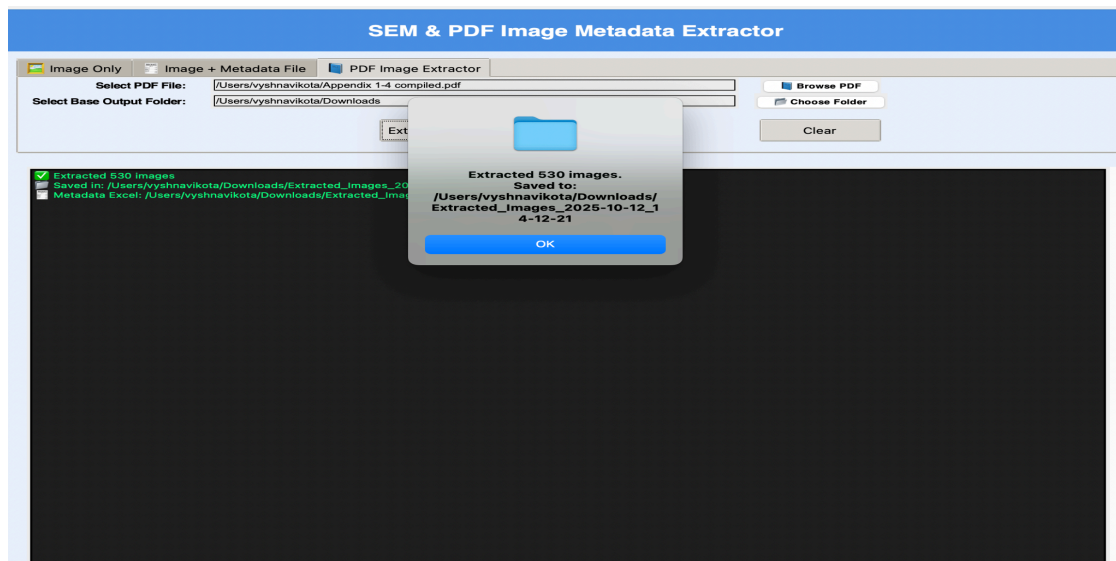
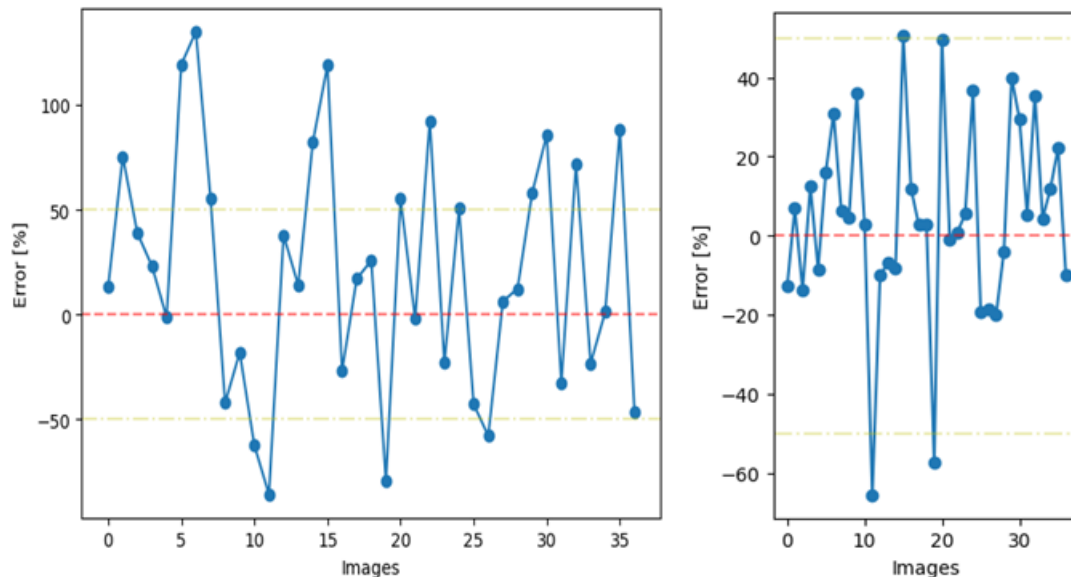


Figure 9: PDF Image Extractor

Overall, the results confirm that the developed application reliably extracts and standardizes metadata across heterogeneous formats, enabling consistent analysis, reproducibility, and streamlined data management in microscopy research.

## 2.2.4 SEM Scale Prediction Model

The models were evaluated and we discovered that the patch solution increased consistency greatly (compared to making predictions for compressed images).



**Figure 10: Before (Left) vs After (Right) Applying Patch Solution**

The mean absolute percentage error decreased from 55.0% to 18.4%, and the standard deviation of percentage errors decreased from 64.1% to 25.0%.

For the long-former solution, a chunk solution[4] implementation was attempted but it was too memory-intensive with the current resources. However, it was noted that the computation speed was significantly faster due to less computations for attention layers.

## 3.1 Discussion

In this section, the discussion will focus on the neural network model, as the other components were implemented with a fixed method from the start.

The model is a neural network model, which is very difficult for humans to interpret. However, this was acceptable as we did not require high explainability for this task, as the use case for this model was when the information was completely missing and the only alternative was expert knowledge. With the model predictions, researchers could set magnification based on

model prediction and make minor adjustments when reproducing analyses. This estimation will not be 100% accurate, but provides a reasonable estimation/starting point.

For the model, three different solutions were proposed and tested, with the findings below:

### Pros and Cons for Different Models

|      | Compress Image            | Patch Solution                              | Long-former Solution                |
|------|---------------------------|---------------------------------------------|-------------------------------------|
| Pros | Making fast prediction    | Making better predictions                   | Could make much better prediction   |
| Cons | Low in prediction quality | Spend more computing resource in prediction | Much more complicated to implement. |

Due to the limitations to size of the training dataset (~300 images), this model may not generalize well. Also, the entire training dataset is in the same resolution, meaning the model may not generalize to images of other resolutions. This problem could be tackled in the future with an increase in training samples.

With the current resources, we have developed a model which has good enough performance for usage, after training with enough data. Also, it is possible to use this model for other tasks related to mineral ore research (like predicting mineral ore type given SEM image). We also explored the possibility for a better solution given more resources.

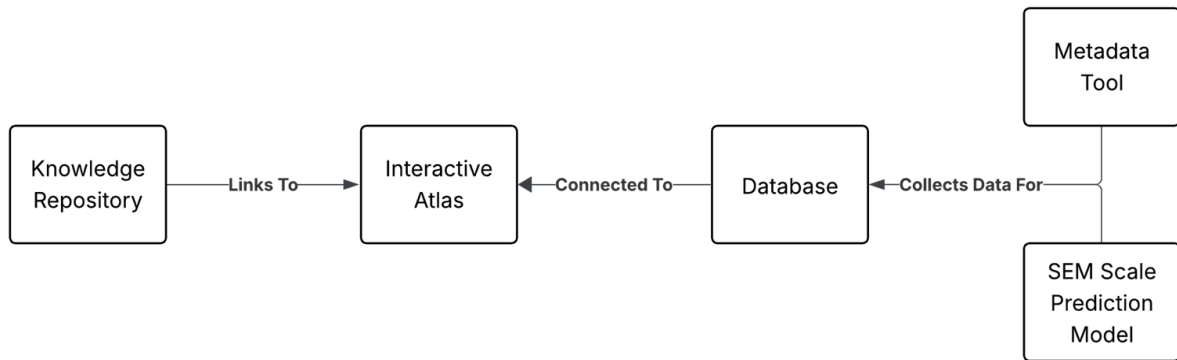
In the future, an implementation of the tvn solution[4] could potentially result in a better model. We could generalize the scale prediction for other sorts of images (e.g. Optical Microscopy images), and other tasks (e.g. predicting instrument type), if we have more labeled data for training.

## 4.1 Conclusion

The delivered components link together to form a solid foundation of a comprehensive system, helping our clients fill their need for a centralized knowledge repository to improve the accessibility and quality of shared knowledge and data, as well as providing additional tools to further improve the quality and efficiency of geoscience research.

In the future, the knowledge repository could be expanded to host a bigger variety of training modules. With PostGIS, the interactive data atlas could also be adapted to perform more advanced visualisations with spatial queries. Hosting the entire system remotely would also greatly improve accessibility within the CMCA. For the metadata tool, more improvements can

be done to handle edge cases and output from more instruments. For the SEM scale prediction model, an increase in training data sample size and computational resources would also build on the current foundation nicely to develop an even better model.



**Figure 11: Schematic Representation of Project**

## 5.1 References

[1] Australian Government, "Digital Atlas of Australia," *Geoscience Australia*, 2025. [Online].

Available: <https://digital.atlas.gov.au>

[2] M. Huisman, M. Hammer, A. Rigano, U. Boehm, J. J. Chambers, N. Gaudreault, A. J. North, J. A. Pimentel, D. Sudar, P. Bajcsy, C. M. Brown, A. D. Corbett, O. Faklaris, J. Lacoste, A. Laude, G. Nelson, R. Nitschke, D. Grunwald, and C. Strambio-De-Castillia, "A perspective on microscopy metadata: data provenance and quality control," *Nature Methods*, vol. 19, no. 11, pp. 1423–1436, 2022, doi: 10.1038/s41592-022-01610-7.

[3] A. Dosovitskiy, L. Beyer, A. Kolesnikov, D. Weissenborn, X. Zhai, T. Unterthiner, M. Dehghani, M. Minderer, G. Heigold, S. Gelly, J. Uszkoreit, and N. Houlsby, "An image is worth 16×16 words: Transformers for image recognition at scale," *arXiv preprint arXiv:2010.11929*, 2020, doi: 10.48550/arxiv.2010.11929

[4] I. Beltagy, M. E. Peters, and A. Cohan, "Longformer: The long-document transformer," *arXiv preprint arXiv:2004.05150*, 2020, doi: 10.48550/arxiv.2004.05150

## 6.1 Appendix

### Task allocation + Contributions of each member

| Member Name                | Student Number | Task for the project                           | Skills contributed to the team and Project                                                                                            |
|----------------------------|----------------|------------------------------------------------|---------------------------------------------------------------------------------------------------------------------------------------|
| Nick Chong                 | 24318278       | Repository+DataAtlas (Map Interface Component) | Database design, Gitbook, Programming, problem solving, communication, team coordination, Report writing                              |
| Dency Pothaparambil Daison | 24544271       | Repository+Data Atlas (Database Component)     | Database design and implementation, Gitbook, Programming, Problem solving, communication, Report writing                              |
| Vyshnavi Kota              | 24418335       | Metadata Tool                                  | Data Analysis, Data Transformation and Extraction, Tkinter, Python Programming, Problem solving, Github Communication, Report writing |

|           |          |                            |                                                                                                                                                  |
|-----------|----------|----------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------|
| Frank Tan | 24478725 | Sem Image Prediction Model | Neural Network Model design and implementation, Programming , Problem solving, communication, TensorFlow, paper search & reading, Report writing |
|-----------|----------|----------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------|

## Team Github Repository

[https://github.com/vyshnavichowdary514-code/CITS5553\\_CapstoneDataScience\\_Group4.git](https://github.com/vyshnavichowdary514-code/CITS5553_CapstoneDataScience_Group4.git)